

REPORT ON HYPERPARAMETER OPTIMIZATION

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Dataset	Cats vs. Dogs Filtered Dataset
Framework	TensorFlow / KerasTuner Hyperband

1. Summary

The report describes the automated hyperparameter optimization of a Convolutional Neural Network (CNN) for the purpose of classifying images of cats and dogs. KerasTuner's Hyperband method was employed in order to search the hyperparameter space that had been defined. The best trial among those conducted reached a validation accuracy of 87.00%. The chosen configuration was then retrained for a maximum of 20 epochs, using early stopping; training was halted after 7 epochs and the highest validation accuracy obtained during this retraining phase was 85.33%.

The manual CNN achieved a validation accuracy of 85.52%. Therefore, the best Hyperband trial was 1.48 percentage points higher than the manual CNN, whereas the final retrained tuned model was 0.19 percentage points lower. The two tuned values must be reported separately since the best trial score and the later retraining run represent different measurements.

2. Experimental Setup

The experiment made use of the Cats and Dogs dataset, which was prepared as 150×150 RGB images, and the notebook contained 6,000 images. KerasTuner employed Hyperband to optimize validation accuracy (val_accuracy), setting the maximum number of epochs per trial to 10 and the factor to 3; early stopping was used to monitor validation loss with a patience value of 3 and including restore_best_weights=True.

3. Hyperparameter Search Space

Hyperparameter	Search Range / Choices	Best Value
Number of convolutional layers	2–4	3
Filters – Layer 1	32, 64, 128	32
Filters – Layer 2	32, 64, 128	128
Filters – Layer 3	32, 64, 128	128
Dense units	128–512, step 128	128
Dropout	0.2–0.5, step 0.1	0.3
Learning rate	0.01, 0.001, 0.0001	0.0001

4. KerasTuner Search Results

The Hyperband search involved 30 trials, and trial 0024 achieved the highest validation accuracy, which was 0.8700000047683716 and was stated as 87.00%. The most successful trial involved three convolutional layers having 32, 128, and 128 filters, 128 dense units, a dropout rate of 0.30, and a learning rate of 0.0001.

Model / Result	Validation Accuracy	Difference vs. Manual CNN
Manual CNN	85.52%	—
KerasTuner Best Trial	87.00%	+1.48 percentage points
Final Retrained Tuned Model	85.33%	−0.19 percentage points

5. Selected CNN Architecture

The best model consisted of three Conv2D-MaxPooling blocks, after which came a Flatten layer, a dense layer with 128 units, 30% dropout, and a single sigmoid output neuron. It used the Adam algorithm with a learning rate of 0.0001 and had binary crossentropy as the loss function.

- Input: $150 \times 150 \times 3$ RGB
- Conv2D: 32 filters $\rightarrow$ MaxPooling2D
- Conv2D: 128 filters $\rightarrow$ MaxPooling2D
- Conv2D: 128 filters $\rightarrow$ MaxPooling2D
- Flatten $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.30) $\rightarrow$ Dense(1, sigmoid)
- Total trainable parameters: 4,920,705 (18.77 MB)

6. Final Retraining

The model which performed best was obtained by using `tuner.get_best_models(num_models=1)`. It was retrained with a 10% validation split, a batch size of 32, and a maximum of 20 epochs, using the same early-stopping callback as before. Training ceased after 7 epochs had been reached. The highest training accuracy achieved was 88.50%, and the highest validation accuracy achieved was 85.33%.

7. Manual CNN vs. Automated Tuning

The manual CNN obtained a validation accuracy of 85.52 per cent. The best trial from KerasTuner reached 87.00 per cent, which was an improvement of 1.48 percentage points. The independently retrained tuned model achieved 85.33 per cent, that being 0.19 percentage points less than the manual CNN.

The fact that this result does not indicate that Hyperband has failed should be noted, since 87.00% is the highest score achieved during the search, while 85.33% is the score that was obtained when the chosen configuration was trained once more. The small differences are due to the stochastic nature of neural-network training.

8. Interpretation of the Selected Hyperparameters

The search selected three convolutional layers, allowing the network to learn progressively more complex image features. The $32 \rightarrow 128 \rightarrow 128$ filter progression gives later convolutional layers more feature capacity. The 0.0001 learning rate provides smaller weight updates, while 0.30 dropout provides regularization. The 128-unit dense layer supplies classification capacity without using the largest dense configuration in the search space.

9. Discussion

The experiment demonstrates that automated tuning is able to systematically assess different CNN configurations rather than depending solely on manual trial and error; Hyperband found a competitive configuration which achieved 87.00% validation accuracy during the search.

The result of the final retraining was 85.33%, which is very near to that of the manual CNN of 85.52%. The experiment thus shows that a tuned model and a systematic optimisation procedure are competitive, but the available results do not allow one to claim a large or consistent improvement upon the manual CNN.

10. Conclusion

KerasTuner's Hyperband algorithm was able to search the hyperparameter space for the CNN used in the Cats and Dogs classification problem. The best trial obtained a validation accuracy of 87.00% by using 3 convolutional layers, 32/128/128 filters, 128 dense units, a dropout rate of 0.30, and a learning rate of 0.0001. Following retraining, the chosen model reached a validation accuracy of 85.33% and stopped at epoch 7 as a result of early stopping.

When compared to the manual CNN result of 85.52%, the best Hyperband trial achieved an improvement of +1.48 percentage points, whereas the final retrained model had a difference of -0.19 percentage points. Altogether, automated tuning offered a systematic method for identifying a competitive CNN configuration.